Transmission System Design Report

I. Rulebook Constraints

A. Transmission Constraints

As Per the Indian Karting Race (IKR) 2025 rulebook, these constraints apply:

- The drivetrain must be of rear-wheel drive configuration only.
- Teams are allowed to utilize chain and sprocket, and gearbox mechanisms.
- There is no restriction on sprocket ratios.
- The transmission system can include a direct axle.
- The entire transmission assembly must be securely mounted.
- A proper shielding is required to cover the moving parts of the transmission.

B. Engine Constraints

- Only single-cylinder, four-stroke engines are permitted.
- Engines may be air-cooled.
- The maximum allowable engine displacement is 165cc.
- The fuel type is limited to petrol.
- Engine tuning should not violate any performance limiters.
- Teams are not allowed to alter the standardized governor setting after inspection.
- Each team must produce a valid purchase bill and recent servicing proof.
- Reuse of the same engine is allowed for up to two consecutive events.

II. Force Consideration

In designing the transmission system, several forces must be considered:

A. Force Produced by the Engine

Power (kW) can be calculated as:

$$P = \frac{T \cdot \omega}{1000}$$

where T is torque (Nm) and ω is angular velocity (rad/s).

B. Rolling Friction

Rolling resistance is:

$$F_{\text{rf}} = \mu_{\text{rf}} M g$$

where μ_{rf} is the coefficient of rolling resistance.

C. Air Drag

Air drag force:

$$F_{\text{air drag}} = \frac{1}{2} \rho C_d A v^2$$

where C_d is drag coefficient, A is frontal area, ρ is air density, and v is velocity.

D. Effective Force

$$F_{\text{effective}} = F_{\text{engine}} - F_{\text{rolling friction}} - F_{\text{air drag}}$$

III. Working Principle

The go-kart's transmission system transfers the engine's torque to the rear wheels through a chain and sprocket. The sequence is:

$$\text{Engine output shaft} \rightarrow \text{Driver sprocket} \rightarrow \text{Chain} \rightarrow \text{Driven sprocket} \rightarrow \text{Rear axle} \rightarrow \text{Wheels}$$

IV. Design Constraints

The driver sprocket is positioned close to the pillow block bearing to minimize bending stresses. A sprocket combination of 13 teeth (driver) and 44 teeth (driven) is used, with a gear ratio of approximately 3.38.

V. Component Selection

A. Engine

TVS RTR 160 2V engine: 13.85 Nm @ 7000 RPM and 16.04 PS (11.79 kW) at 8750 RPM.

B. Sprockets

13-tooth driver and 44-tooth driven sprocket. Sprocket ratio approximately 3.38:1.

C. Rear Axle

Fabricated from AISI 4140 steel for high strength and fatigue resistance.

D. Justification

Components were selected for optimal torque delivery, acceleration, top speed, and mechanical toughness.

E. Analysis of Components

The analysis of the rear shaft was performed considering various loads acting during acceleration, braking, and static conditions. The key forces and resulting torques are described below.

TABLE I: Axle Specifications

Property	Value
Length	1200 mm
Diameter	30 mm
Material	AISI 4140 Steel

1) Driving Torque: The primary torsional load during acceleration is applied at the sprocket mounting location.

$$T_{\text{axle}} = T_{\text{engine}} \times \text{GR} \times \text{FDR} \times \eta$$

where:

- $\eta = 0.88$ (mechanical efficiency)
- FDR = $\frac{44}{13} = 3.385$ (Final Drive Ratio)
- First GR = 2.875 (Gearbox Ratio)
- $T_{\text{engine}} = 13.85$ Nm

Thus,

$$\begin{aligned} T_{\text{axle}} &= 13.85 \times 2.875 \times 3.385 \times 0.88 \\ &= 118.48 \text{ Nm} \quad \text{(Anti-clockwise)} \end{aligned}$$

Applied at the sprocket position, 275 mm from the center.

2) Braking Torque: The braking system applies an opposing torque at the brake disc mounting location.

$$\text{Braking Torque} = 85 \text{ Nm} \quad \text{(Clockwise)}$$

Applied at 275 mm from the center.

3) Vertical Wheel Loads: Vertical forces are induced by the go-kart's weight, mainly on the rear axle bearings.

$$\text{Total Weight} = 180 \times 9.81 = 1765.8 \text{ N}$$

Assuming 60% of the total weight acts on the rear shaft:

$$\begin{aligned} \text{Force at each bearing} &= \frac{0.60 \times 1765.8}{2} \\ &= 556.227 \text{ N} \approx 560 \text{ N} \end{aligned}$$

Applied downward at bearing locations, 330 mm away from the center.

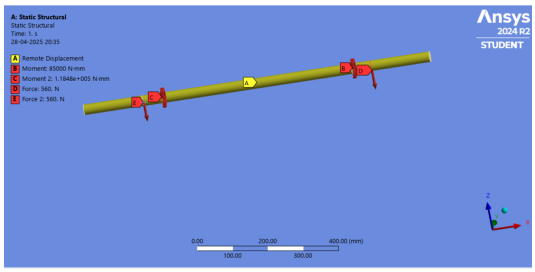

Fig. 1: Rear Shaft Force Conditions

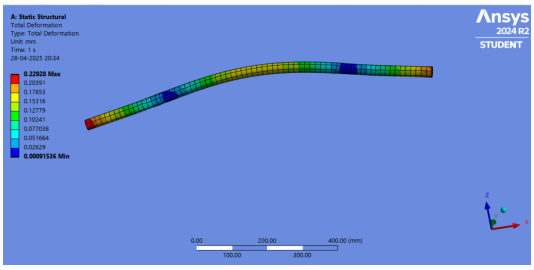

Fig. 2: Rear Shaft Total Deformation

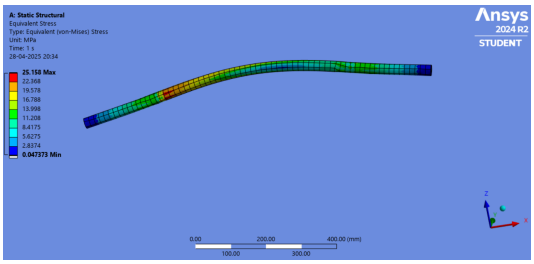

Fig. 3: Rear Shaft Equivalent Stress

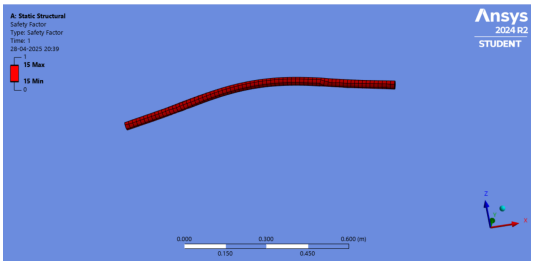

Fig. 4: Rear Shaft Factor of Safety

From the analysis it can be concluded that the selected dimensions of the shaft sufficiently meet the requirements of the setup and can perform under stress

VI. Formulation and Calculation

A. Equations

- Final output power:

$$P_F = P_{\max} \cdot \eta_t$$

- Torque by Engine:

$$T_E = \frac{P_{\max} \cdot 60}{2\pi N_{\max}}$$

- Torque to wheels:

$$T_{\text{wi}} = T_E \cdot G_i \cdot G_{E-S} \cdot \eta_t$$

- Force by engine:

$$F_{\text{engine}(i)} = \frac{T_{\text{wi}}}{r_w}$$

- Air drag force:

$$F_{\text{air drag}(i)} = \frac{1}{2}\rho C_d A v_i^2$$

- Rolling friction:

$$F_{\text{rolling friction}} = \mu_{\text{rf}} M g$$

- Effective force:

$$F_{\text{effective}(i)} = F_{\text{engine}(i)} - F_{\text{rolling friction}} - F_{\text{air drag}(i)}$$

- Average acceleration:

$$a_i = \frac{F_{\text{effective}(i)}}{M}$$

- Time from velocity U to V:

$$T_{U-V} = \sum \frac{v_{fi} - v_{si}}{a_i} \times \frac{5}{18}$$

- Top speed ignoring air drag:

$$v_{\text{top}} \leq \frac{r \cdot 2\pi}{60} \cdot \frac{N_{\text{max}}}{G_5 \cdot G_{E-S}} \cdot \frac{18}{5}$$

- Top speed considering air drag:

$$v_{\text{top}} \leq \sqrt{\frac{F_{\text{engine}(5)} - F_{\text{rolling friction}}}{\frac{1}{2}\rho C_d A}}$$

$$\text{(For air drag)} \quad F_{\text{effective}} = 0$$

B. Parameters

- Max Power: 11.797 kW
- Max RPM: 8750
- Transmission efficiency: 88%
- Mass: 180 kg
- Wheel Diameter: 10 in = 0.2794 m
- Gravity: $9.81\, m/s^2$
- Rolling friction coefficient: 0.012
- Air drag coefficient: 0.6
- Air density: $1.2\, kg/m^3$
- Frontal Area: $0.8\, m^2$

C. Tables and Graphs

TABLE II: Gearbox Ratios and Operating Velocities

Gear	Ratio	Initial Vel	Final Vel	Average Vel
		(km/h)		
1	2.875	0.0	14.0	7.0
2	2.000	14.0	25.0	19.5
3	1.480	25.0	47.3	36.2
4	1.185	47.3	75.5	61.4
5	1.050	75.5	109.3	92.4

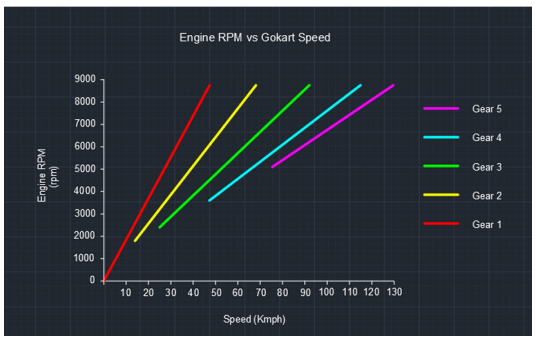

Fig. 5: Engine RPM vs Go-kart Speed across different gears.

TABLE III: Torques, Forces, and Accelerations

Gear	Torque (Nm)	F_{eng} (N)	F_{drag} (N)	F_{roll} (N)	F_{eff} (N)	Accel. (m/s^2)
1	110.3	789.3	1.1	21.2	767.0	4.3
2	76.7	549.1	8.4	21.2	519.5	2.9
3	56.8	409.3	29.0	21.2	359.1	2.0
4	45.4	325.3	83.8	21.2	220.3	1.2
5	40.3	288.3	189.7	21.2	77.4	0.43

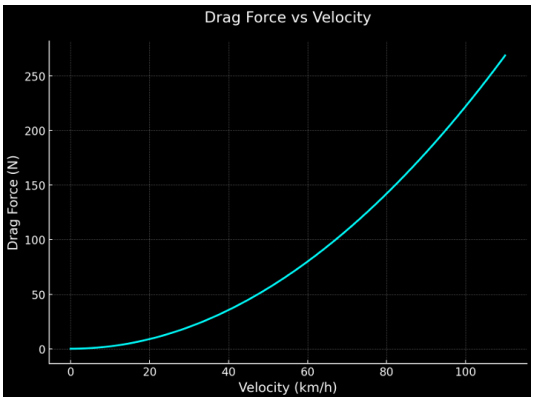

Fig. 6: Drag Force vs Velocity for the Go-kart.

VII. Results

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- Final output power (P_p) = 10.381 kW
- Maximum torque of engine (T_{max}) = 13.85 N·m @ 7000 RPM (Given)
- Torque outputs at the wheels:
  - At 1st Gear – 110.3 N·m
  - At 2nd Gear – 76.7 N·m
  - At 3rd Gear – 56.8 N·m
  - At 4th Gear – 45.4 N·m
  - At 5th Gear – 40.3 N·m
- Average acceleration at each gear:
  - a_1 = 4.3 m/s^2
  - a_2 = 2.9 m/s^2
  - a_3 = 2.0 m/s^2
  - a_4 = 1.2 m/s^2
  - a_5 = 0.43 m/s^2
- Average initial acceleration (a_{0-40}) = 3.52 m/s^2
- Top speed (v_{top}) = 109.6 km/h